Four Pitfalls in Measuring Container and Observability Overhead: A Replicated Case Study

Vinayak Tyagi
Department of Computer Science and Engineering
Manipal University Jaipur, Jaipur, India
vinayaktyagi.ed@gmail.com

***Abstract*—We measured a small containerized pipeline (Docker, GitHub Actions, Prometheus and Grafana) against uncontainerized and unmonitored baselines, then repeated the study on the same hardware after reinstalling the operating system. Four measurement pitfalls changed what the data appeared to show: (P1) *zero pages are not memory pressure*: a zero-filled stress test stopped triggering OOM kills on the replication kernel because its transparent-huge-page shrinker, enabled by a non-default distribution setting, reclaimed the zeroed pages; (P2) *the load client measures the socket layer*: a client opening a new connection per request measured ephemeral-port exhaustion in `docker-proxy`, not the application; (P3) *the sampling window outlasts the load*: `docker stats` samples longer than a 0.11 s load produced an apparent CPU-contention effect that kernel counters did not confirm; (P4) *absolute numbers do not travel*: most values moved by 12–173% across the reinstall, and by 27% between two later sessions after routine package updates. P2–P4 are known practice; P1 is the one we did not find documented. What held: the ordering of deployment times and the cost of re-running build and test on every CI/CD deployment (both follow from what each configuration does), recovery time after a container kill, and an OOM kill when memory is resident. The first campaign's explanation of a throughput drop as CPU contention between monitoring and application did not hold and is withdrawn: with a stable client the monitoring stack used about 0.03 cores and cost at worst about 1.4% of throughput, with no 95% interval reaching a loss above 2.3%, and the cost did not grow under background CPU load. An interleaved re-run of the original experiment, with an identical-configuration control, found no monitoring effect but a downward drift of about 16% over 11 minutes, enough to produce the original drop under a fixed case order. For each pitfall we give the symptom, mechanism, evidence and a check that prevents it. Raw data and a one-command rebuild of every table and figure are published at https://github.com/vinayaktyagi10/cicd-monitoring-study.**

***Index Terms*—performance measurement, reproducibility, containers, Docker, CI/CD, Prometheus, observability overhead, transparent huge pages**

I. Introduction

Evaluations of containerized deployment, CI/CD automation and monitoring stacks usually report one measurement campaign on one host and treat the resulting numbers as properties of the tools. We set out to do the same for a small pipeline: a purpose-built FastAPI service, packaged with Docker, built and tested by a GitHub Actions workflow, and monitored by Prometheus and Grafana. We measured deployment time, resource use, throughput, and how quickly the monitoring detects induced CPU saturation, memory exhaustion and container failure, against an uncontainerized process and an unmonitored container. The first campaign told a clean story, including a counter-intuitive one: adding monitoring containers lowered both the application's CPU reading and its throughput, which we read as CPU contention between the monitoring stack and the application.

That story did not survive replication. Repeating the experiments on the same hardware after the operating system was reinstalled moved most absolute numbers outside their original confidence intervals, a memory-stress test that had OOM-killed the container in every run stopped killing it at all, and the contention effect disappeared. Tracing each change to a cause turned up four ways a container or observability experiment can report something it did not measure:

- **P1. Zero pages are not memory pressure.** On the replication kernel, with transparent huge pages enabled, the kernel freed mostly-zero huge pages, so a container holding zeroed memory reached its limit without being OOM-killed (Section IX-A).
- **P2. The load client measures the socket layer.** A client that opens a new TCP connection per request exhausts ephemeral ports within seconds, and throughput then tracks `docker-proxy` system time, not the application (Section IX-B).
- **P3. The sampling window outlasts the load.** A `docker stats` reading that mostly misses a 0.11 s burst makes CPU appear to fall as monitoring is added (Section IX-C).
- **P4. Absolute numbers do not travel.** On the same hardware, deployment times, throughput and detection latencies moved by 12–173% after a reinstall (Section IX-D).

P2, P3 and P4 are known measurement practice (port exhaustion under per-request connections, sampling windows that miss short loads, environment sensitivity of absolute numbers), and we do not present them as discoveries. P1 is the one we did not find documented in the work we reviewed: it comes from how transparent huge pages, the kernel's underused-page shrinker and a distribution's non-default setting interact, and it turns a zero-filled memory stress test into one that applies no pressure. What we add, for all four, is a record of how each produced a plausible, wrong or fragile result in a study that was otherwise logged and repeated carefully, the diagnostic that exposed it, and a cheap check that prevents it. A small literature audit (Section III-B) finds that the settings behind

P1 and P4 are rarely reported. The pipeline is the case study, not the contribution.

A. Research Questions

The case study asks three questions about the pipeline and one about the measurements themselves.

- **RQ1.** How does containerization affect deployment time and resource use, and what does CI/CD automation add?
- **RQ2.** How quickly does Prometheus detect induced CPU saturation and container failure?
- **RQ3.** What does running Prometheus and Grafana alongside the application cost?
- **RQ4.** Which of these findings survive a change of software environment on the same hardware, and which measurement choices explain the ones that do not?

II. Contributions

1) **Four measurement pitfalls, each with mechanism and evidence; P1 is the substantive one.** For P1 we show with a crossed diagnostic that disabling transparent huge pages for the one process restores the OOM kill (Table IX), and with a host-wide sweep in a guest that the escape needs a non-default `max_ptes_none` (Table XI). For P2 we show throughput collapsing as TIME_WAIT sockets pass the ephemeral port range (Fig. 11). For P3 we show the bimodal CPU reading and its disappearance under kernel-counter measurement. For P4 we classify every finding as reproducible or environment-dependent (Table XIII). Table I gives the check that prevents each one.
2) **A retracted claim, reported as retracted.** The CPU-contention explanation from the first campaign is tested against five of its own predictions and withdrawn (Section VIII-I); what remains is a bound on monitoring overhead: about 1.4% of throughput at worst, with 95% intervals reaching 2.3%.
3) **Case-study findings that held, and how much they are worth.** Deployment ordering, the CI/CD build-and-test cost, recovery time after a container kill, peak CPU under stress, the Prometheus and Grafana CPU draw, and the OOM kill with resident memory all reproduced. The first two follow from what each configuration does and are weak evidence about the tools; the rest are measurements of one small service on one host.
4) **A traceable artifact.** Raw per-run data, experiment and analysis scripts, and a one-command rebuild of every table, statistic and figure from that raw data (`make reproduce`, no Docker needed), with every number quoted in the text checked against the data. Re-collecting data is repeatable but not deterministic (Section XI).

III. Related Work

This section covers the work closest to each tool in the case study and the two methodology studies we position the pitfalls against. It is a targeted review, not a systematic one. Table II summarizes prior evaluations of the tools this case study uses.

Felter et al. [1] remains the most-cited apples-to-apples comparison of native execution, Docker containers, and KVM virtual machines, finding containers add near-zero CPU/memory overhead relative to native but that both containers and VMs pay a cost on I/O-bound workloads. Lingayat et al. [2] and Mavridis and Karatza [3] extend this line to deployment/startup time and to containers nested inside VMs respectively, but both stop at the container layer: neither introduces a CI/CD pipeline or a monitoring stack, so their measurements cannot speak to RQ2–RQ3. Ekanayaka et al. [4] is the closest system to the one built here — *KubFlow* integrates Git, Jenkins, ArgoCD, Terraform, Docker, Kubernetes, Prometheus, and Grafana into a single DevOps infrastructure — but its evaluation is a pass/fail checklist of eight qualitative test scenarios plus subjective colleague ratings, not a quantitative, repeated-measures comparison against a non-monitored or non-automated baseline.

Three further works bear directly on individual pillars. Dinga et al. [5] empirically evaluate the energy and performance overhead of four monitoring tools, Prometheus among them, on Docker-based systems and report that the overhead depends on the tool and on the frequency and workload conditions; this is the closest prior work on monitoring overhead, but it isolates the monitoring tools and does not combine them with a CI/CD pipeline or fault injection. Our Experiment 6, which measures the monitoring containers' CPU from kernel counters, is complementary: it found $\approx$0.03 cores for cAdvisor, Prometheus and Grafana together on this host, which is compatible with a tool-dependent overhead but is a single configuration and not a comparison across tools. Hilton et al. [6] study the usage, costs and benefits of continuous integration across 34,544 open-source GitHub projects, which motivates measuring the cost of the CI step itself, as the Docker+CI/CD arm does. Forsgren, Humble and Kim [7] establish time-to-restore and change-failure rate as standard software-delivery performance measures; our MTTD and MTTR are the container-level analogues, measured in seconds on a single service rather than as organization-level outcomes. Container background is covered by Bernstein [8].

Earlier container work includes Xavier et al. [9], who found container-based virtualization close to native for HPC workloads; like Felter et al., it measures the container layer only.

Several studies address measurement methodology rather than any one tool. Georges et al. [10] showed that common Java benchmarking practice can mislead because the data analysis is not statistically rigorous, and advocated confidence intervals over best-of-n summaries. Kalibera and Jones [11] derived how to allocate repetitions across levels of a non-deterministic experiment for a given time budget, and Hoefler and Belli [12] surveyed 120 papers from three conferences and found reporting and experimental-design practice lacking. Papadopoulos et al. [13] proposed eight principles for measuring and reporting cloud performance and found that few published studies follow them. Abedi and Brecht [14] argue that randomized multiple interleaved trials support repeatable

TABLE I
MEASUREMENT PITFALLS FOUND IN THIS STUDY: WHAT EACH LOOKED LIKE, WHY IT HAPPENED, HOW IT WAS EXPOSED, AND A CHECK THAT PREVENTS IT.

Pitfall	Symptom here	Mechanism	How it was exposed	Check that prevents it
P1 Zero-filled memory stress	0/20 OOM kills on the replication host, 20/20 originally	THP shrinker splits mostly-zero huge pages and frees them under cgroup pressure	`memory.events` at the limit with no OOM kill; THP × fill diagnostic, stride probe and guest sweep (Tables IX, X, XI)	Write non-zero data to every page; report kernel, THP policy, shrinker, `max_ptes_none` and fill pattern
P2 Per-request load client	Throughput varied three-fold between runs and tracked docker-proxy CPU	New TCP connection per request exhausts ephemeral ports; TIME_WAIT and docker-proxy system time dominate	Per-second TIME_WAIT and proxy CPU series (Fig. 11); second client with keep-alive	Use keep-alive connections; record TIME_WAIT, client and proxy CPU; check within-case variation
P3 Sampling window longer than the load	Application CPU% fell as monitoring was added	A 0.11 s burst mostly falls between `docker stats` samples, so many runs read idle	Bimodal per-run readings; kernel counters over the exact load window	Use a fixed-duration load; read cgroup `cpu.stat` at the window edges
P4 Absolute numbers across environments	Deployment, throughput and detection times moved 12–173% after a reinstall; 27% between two later sessions	Kernel, Docker version and distribution defaults changed on the same hardware	Same-hardware replication with a verdict rule fixed in advance (Table XIII); power-profile test explains a few percent of the 27%	Record package versions and power state per run; pin versions; randomize case order; run an A/A comparison for the noise floor

TABLE II
RELATED WORK SUMMARY

Paper	**CI/CD**	**Containers**	**Monitoring**	**Metrics**	**Scope (relative to this study)**
Felter et al. 2015 [1]	No	Docker vs. KVM	No	CPU, memory, network, disk microbenchmarks; MySQL/Redis throughput	Isolates container vs. VM overhead only; no CI/CD, no monitoring stack, no failure-detection measurement.
Lingayat et al. 2018 [2]	No	Docker vs. bare metal and VM	No	Container startup time	Single metric (startup time); no automation pipeline, no monitoring, no fault injection.
Mavridis & Karatza 2017 [3]	No	Docker nested in VM	No	Standard container benchmarks	Studies an orthogonal deployment topology (container-in-VM); no CI/CD or monitoring integration.
Ekanayaka et al. 2023 [4]	Jenkins, ArgoCD	Kubernetes	Prometheus, Grafana	Pass/fail test scenarios; subjective 1–5 feature ratings	Same three pillars as this study, but evaluation is qualitative/subjective, not a controlled quantitative comparison against a non-monitored or non-automated baseline.
This study	GitHub Actions	Docker Compose	Prometheus, Grafana	Deployment time, reliability, CPU/mem utilization, throughput, latency, MTTD, MTTR; n=20 per configuration where feasible	—

comparisons when conditions outside the experimenter's control vary, and Laaber et al. [15] found that running test and control on the same instances in randomized order detects slowdowns of 10% or less. Variability itself is documented for public clouds by Leitner and Cito [16] and for big-data workloads on cloud networks by Uta et al. [17]; Curtsinger and Berger [18] attack measurement bias from memory layout by randomizing it. Two further studies are the closest in spirit to ours. Mytkowicz et al. [19] showed that setup details that look irrelevant, such as the size of the UNIX environment and the link order, bias benchmark results enough to reverse a conclusion, and that this bias goes unnoticed unless the setup is varied. Maricq et al. [20] measured performance variability across many nominally identical servers of a shared testbed and gave guidance on how many repetitions a claim needs and on using nonparametric confidence intervals. Both concern general benchmarking: the benchmark binary, the hardware and run-to-run variance. The interleaved-trial and A/A designs of Abedi and Brecht and of Laaber et al. are what Experiment 2 lacked and what Section VIII-I adds.

A. Positioning

Each tool evaluation above reports a single measurement campaign, and none repeats its measurements in a second software environment or reports a result that changed when it was repeated. Dinga et al. [5] is the closest in subject and shows that monitoring overhead depends on the tool and the workload; we do not compare tools, and our overhead bound is for one configuration on one host. We make no claim that combining containers, CI/CD and monitoring is itself a research question, nor that the pitfalls in Section IX are individually unknown. Our claim is narrower: in one replicated study, each of them produced a wrong or fragile result that a single campaign would have reported as a finding, and each is detectable with a cheap check. Relative to the methodology studies, P1–P3 arise in container and observability measurement: the kernel's handling of a container's memory, the container runtime's port proxy and the monitoring tool's sampling window, not the benchmark binary or the hardware. Of these only P1 is, as far as we found, undocumented; P2 and P3 are container-specific instances of port exhaustion and sampling-window mismatch, which are well known in general. A small audit of published studies (Section III-B) covers only what they report, not how often they commit any of the four. P4 is the lesson of Mytkowicz et al. and Maricq et al. at a different level: on unchanged hardware, a change in the software environment moved results further than a single campaign's intervals would suggest. We follow Maricq et al. only in part: our main intervals are t-based, with rank tests and bootstrap intervals as checks (Section VII).

B. A Small Audit of What Published Studies Report

To see whether the settings behind P1 and P4 are normally reported, we read the setup description of published container and observability overhead studies. The sample is not systematic. We searched the web and arXiv with queries on container and tracing overhead, applied a fixed inclusion rule (an empirical study of Docker or other container overhead, or of monitoring or tracing overhead on containerized or microservice workloads; surveys, position papers, simulation-only studies and papers with no measured overhead excluded), and kept 45 papers from 2013 to 2025 that we could read in full. Each paper was scanned for the fields below and the matches were checked by reading the setup text, or the page image where the text did not extract. One model did the searching, scanning and checking, so the checks are not independent, and a field the scan did not find may still appear in text we missed; the counts are therefore lower bounds on what is reported.

Of the 45 papers, 12 state the kernel version of their own test hosts, and 3 more mention a kernel only in a way that does not identify the host's. None states the transparent huge page policy. None randomizes or interleaves the order of configurations; one describes randomized run order only as background. One runs an identical-configuration control, and that paper is itself about benchmarking methodology; it also found that Docker containers produced more false positives than the other isolation setups in such controls [21]. Another measured the same Docker workload on three operating systems on identical hardware and found the platform mattered [22], which is the P4 setting at a larger scale.

This audit shows only that the P1 and P4 settings are rarely reported. It does not show how often a study committed P1, P2 or P3: most of the 45 are benchmark-suite studies that inject no memory pressure, so the pitfalls rarely apply and the denominators are too small to estimate a rate.

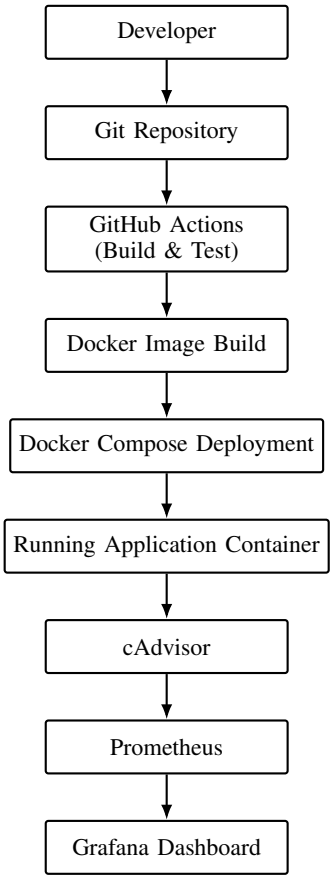


Fig. 1. Pipeline architecture: push, build and test, image, and deployment beside the observability stack.

IV. System Under Test

The pipeline is the measurement subject, not a proposed design; it uses each tool in its default configuration. Fig. 1 shows its architecture and data flow. A developer pushes to the Git repository, which triggers a GitHub Actions workflow that runs the test suite and, on success, builds and pushes a Docker image. The image is then run under Docker Compose alongside a cAdvisor container-metrics exporter, a Prometheus server that scrapes both the application's native `/metrics` endpoint and cAdvisor, and a Grafana instance provisioned with Prometheus as its data source.

Fig. 2 details the GitHub Actions workflow itself, exactly as defined in the repository's `ci.yml`: a `test` job (checkout, Python 3.12 setup, dependency install, `pytest`) gates a

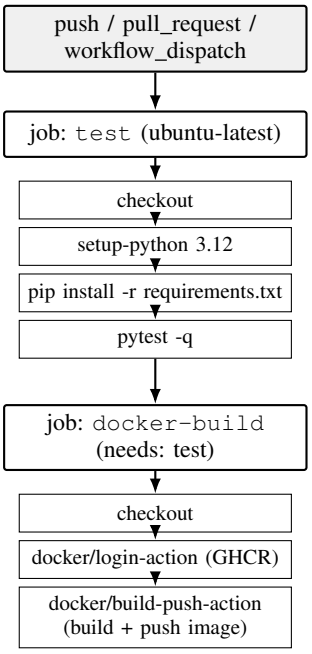


Fig. 2. GitHub Actions workflow (`.github/workflows/ci.yml`). The `docker-build` job only runs if `test` succeeds (`needs: test`); this is the real workflow triggered for the Docker+CI/CD deployment-time measurements in Table VI.

`docker-build` job (checkout, GHCR login, build and push) via `needs: test` — the build only runs if the tests pass. This is the real workflow triggered for every Docker+CI/CD measurement in Table VI, not a simplified stand-in.

Fig. 3 shows the resulting Grafana dashboard under live load: request rate and p95 latency by endpoint, the application's self-reported process CPU% and memory, plus a Prometheus `up` target-health panel (not shown; it reported `up=1`). All panels are auto-provisioned, with no manual dashboard configuration, and update in real time from the same Prometheus instance used throughout the experiments.

Subject application. Rather than an existing multi-service demo application, the study uses a small, hand-built FastAPI service with three endpoints designed as clean experimental levers: `/health` (a near-constant-time route used as the load-test and liveness target), `/cpu` (a configurable-iteration pure-Python busy loop used for CPU stress), and `/memory` (a configurable-size heap allocation, optionally retained, used for memory stress). This choice trades away the multi-service realism of a demo application such as Sock Shop for the ability to attribute every measured effect to a specific, known cause rather than to inter-service scheduling or network confounds. The application exposes Prometheus metrics natively via the `prometheus_client` library — a request counter and latency histogram per endpoint, plus two self-reported process gauges, `app_process_cpu_percent` and `app_process_memory_bytes`, sampled every second from `time.process_time()` and `/proc/self/status` deltas respectively — rather than through a side-car exporter.

TABLE III
EXPERIMENTAL ENVIRONMENT

Parameter	Value
CPU	AMD Ryzen 7 8845HS (8 cores / 16 threads)
RAM	14 GiB
Host OS	Arch Linux (Omarchy), kernel 7.1.9-arch1-2
Docker	29.7.2 (containerd snapshotter)
GitHub Actions	`ubuntu-latest` hosted runners
Prometheus	v2.55.1, 2s scrape/evaluation interval
Grafana	11.2.0
cAdvisor	v0.49.1
Application base image	`python:3.12-slim`
Application framework	FastAPI 0.115.0 / Uvicorn 0.30.6
Load generator	Python script: `ThreadPoolExecutor` + `urllib`, no external binary

V. EXPERIMENTAL SETUP

Table III lists the experimental environment. All experiments ran on a single physical host (no cloud VM); GitHub Actions hosted runners were used only for the CI/CD build-and-test step itself, so that step reflects a genuinely independent, reproducible environment while deployment, load generation, and monitoring remained under full local control.

A load-testing tool such as `hey` or `wrk` was deliberately not installed; a small standard-library Python script (`loadgen.py`) issues concurrent HTTP requests via `ThreadPoolExecutor` and records per-request latency, trading raw throughput ceiling for zero external dependencies and full inspectability.

VI. EXPERIMENTAL SCENARIOS

Five experiments were run, summarized in Table IV, followed during validation by a sixth that re-measures monitoring overhead under controlled load (Section VI-F) and a seventh that repeats that measurement under background CPU load (Section VIII-H), and an interleaved repeat of Experiment 2 (Experiment 2i, Section VIII-I). Each is described in more detail in the corresponding results subsection. A dedicated script per experiment (`exp1_deploy.py` through `exp5_failure_recovery.py`) brings the relevant Docker Compose profile up, executes the scenario, records raw per-run measurements to CSV, and tears the stack down before the next run, so runs are independent of one another. An `aggregate.py` script then computes the mean/standard deviation/min/max summaries reported in Section VIII directly from these raw CSV files.

A. Normal operation and baseline comparison (Experiment 1)

Four configurations are compared, only two of which are baselines proper: *manual* (a raw `uvicorn` process, no container — **Baseline A**) and *Docker* (a locally built and started container, no CI/CD, no monitoring — **Baseline B**). The remaining two are not baselines but a progression toward the full stack: *Docker+CI/CD* (a real GitHub Actions `workflow_dispatch` run — test job then Docker build/push — timed end-to-end, then deployed) is an **intermediate configuration** that isolates what automation alone

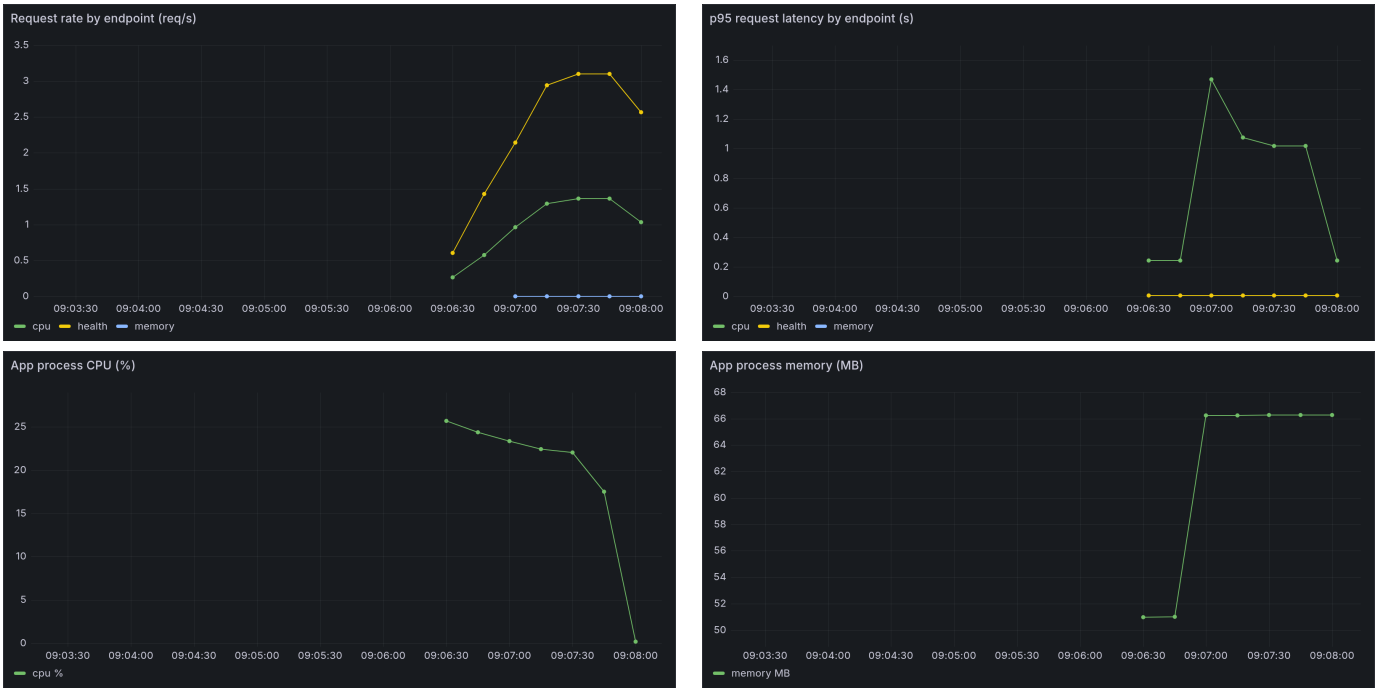


Fig. 3. Grafana panels under a scripted workload (last 5 min): request rate and p95 latency (top), process CPU and memory (bottom). Live screenshots from `capture_grafana.py`, not pixel-reproducible; series in `experiments/raw/grafana_panels.json`.

TABLE IV
EXPERIMENTAL SCENARIOS

#	Scenario	Configurations × runs
1	Deployment time & reliability	manual (20), Docker (20), full pipeline (20), Docker+CI/CD (8)
2	Resource utilization, response time, throughput, monitoring overhead	Docker (20), Docker+Prometheus (20), Docker+Prometheus+Grafana (20)
3	CPU stress detection	20
4	Memory stress / OOM behaviour	20
5	Failure detection & recovery	20
6	Monitoring overhead vs. host CPU vs. throughput (validation)	4 cases (20 each) + pinned arm, two load clients
7	Monitoring overhead under background CPU load (validation)	2 cases × 4 load levels (15 each)
2i	Exp. 2 interleaved with an A/A arm (validation)	4 cases × 20 blocks, shuffled

costs, with monitoring still absent; and *full pipeline* (Docker Compose bringing up the application alongside cAdvisor, Prometheus, and Grafana) is the **full stack**, adding observability on top. This ordering — Baseline A → Baseline B → Docker+CI/CD → full pipeline — lets each of containerization, automation, and observability be attributed its own marginal cost rather than only the combined effect of all three at once. For each run, whatever was previously running is torn down, the configuration is started, and wall-clock time to the first successful health check (and, for the full pipeline, to Prometheus and Grafana also responding) is recorded.

B. Resource utilization and monitoring overhead (Experiment 2)

Three cases isolate the marginal cost of each monitoring layer: Docker only, Docker+Prometheus(+cAdvisor), and Docker+Prometheus+Grafana(+cAdvisor). For each of 20 runs per case, a fixed load (300 concurrent requests to `/health`, concurrency 10) is issued while `docker stats` is polled every second for every container in that case, giving per-container mean CPU% and memory alongside the load test's own throughput, mean/p95 latency, and error rate. The three cases ran in the fixed order Docker, +Prometheus, +Grafana, each on one stack brought up once and loaded 20 times in a row; cases were not interleaved or randomized, so a case is confounded with its position in the session and the 20 runs are repeated loads on one stack instance (Section VIII-I).

C. CPU stress detection (Experiment 3)

Eight threads concurrently flood `/cpu` (an 8-million-iteration pure-Python loop per call) against the Docker+Prometheus case. Detection time is measured as wall-clock time from load start until Prometheus first reports the application's self-reported `app_process_cpu_percent` gauge at or above 60%, polled every 0.5s.

D. Memory stress / OOM behaviour (Experiment 4)

The application container is run directly (not via Compose) with a hard 150 MB memory limit and `--restart on-failure:1`. Repeated `/memory?mb=20&hold=true` calls accumulate retained

allocations until the kernel OOM-kills the container; `docker inspect` gives ground-truth `OOMKilled` and `RestartCount`, captured at the moment of observation since Docker resets `State.OOMKilled` once the restart policy brings the container back up.

E. Application failure and recovery (Experiment 5)

Against the Docker+Prometheus case, the application container is killed (`docker kill`) and Prometheus's `up{job="app"}` series is polled every 0.3s until it reports 0 (detection time). The container is then restarted (`docker start`) and both `/health` and `up{job="app"}==1` are polled until both recover (recovery time).

F. Monitoring overhead under controlled load (Experiment 6)

Added during validation because Experiment 2's ≈0.11 s load window is shorter than one `docker stats` sample. Four cases (Docker, +cAdvisor, +Prometheus, +Grafana) each receive a closed-loop load of 20 s at concurrency 10 against `/health` on a freshly started stack, with case order shuffled within each of 20 blocks. CPU is read from kernel counters at exactly the start and end of the load window: `/proc/stat` for the whole host, each container's cgroup v2 `cpu.stat`, the load generator's own process time, and the `docker-proxy` processes that forward the published port; whatever remains is unattributed background. A CPU-pinned arm places the application, load generator and monitoring containers on disjoint physical cores for the Docker and +Grafana cases. The experiment was run twice with two load clients: the Experiment 2 client, which opens a new TCP connection per request, and a keep-alive client that reuses one HTTP/1.1 connection per worker (`exp6_overhead.py --keepalive`). Raw per-run rows (one per run, with host, per-container and load-generator CPU, throughput and latency) are in `experiments/raw/exp6_overhead*.csv` (four files: both clients, each at one and four load-generator processes); the analysis is `analyze_overhead.py`.

VII. Statistical Analysis

Every interval and test below is computed from the per-run CSV rows, with sample statistics (denominator $n-1$) and a two-sided α=0.05, by `stats_lib.py` (unit-tested against `scipy`) and `validate.py`. Table V lists each method with the reason for choosing it and its main limit. Following Georges et al. [10] and Hoefler and Belli [12], we report intervals and effect sizes first and p-values second: with 20 runs per cell a p-value alone says little about whether a difference matters. The Experiment 2 comparisons in Section VIII quote Cohen's d (`statistical_analysis.py`); everywhere else we use Hedges' g, which differs only by a small-sample correction (under 3% at n=20 per group).

Checking the assumptions. We tested the 30 original-versus-replication metrics of Table XII (including the non-zero-fill Experiment 4 rows). A Shapiro–Wilk test rejects normality in at least one of the two groups for 22 of 30 metrics, so the t-based intervals are approximate for most of them. Welch's test and Mann–Whitney agree on 27 of 30 metrics at α=0.05, which is why we report both. The disagreements are informative. MTTR has equal means (0.7% apart; Welch p=0.52) but a rank test separates the distributions (Cliff's $\delta=-0.80$, Holm-adjusted p=0.0002): the replication's recovery times are tighter, a difference in spread that is practically negligible for a metric near 2.0 s but is not "identical". Prometheus CPU under +Grafana and the non-zero-fill time to OOM also split the two tests.

Multiple comparisons. The inline tests in the Results (the Experiment 2 pairwise comparisons, the Experiment 6 contrasts, the four per-level Experiment 7 penalties and the power-profile test in Section IX-D) are not corrected, and we say so where they appear. Experiment 7 had one test fixed before data collection, the 60% against 0% difference-in-differences; its per-level intervals are descriptive. For the replication comparison we also computed Holm-adjusted p-values across all 30 metrics: 20 of 22 rows marked different keep $p < 0.05$ after the correction. The two that do not are the non-zero-fill OOM rows. There the interval rule says different, while Welch's test finds the two groups indistinguishable (p≈0.33), because every replication run is nearly identical (5 requests, 11.06 s), so the rule's interval collapses to a point. The interval rule and Welch's test agree on 28 of 30 metrics, and the two they split on are exactly these. We keep the rule as stated, since it was fixed in advance, and read those two rows as reproduced in Table XIII.

What we did not do. We did not power the experiments for small effects: the Experiment 6 keep-alive contrasts have 10–20 runs per case, so differences near 1% are at the edge of detection. We analyzed Experiment 6 as independent groups although its runs come in randomized blocks, and a blocked or paired analysis would likely be more sensitive. We did not model dependence between sequential runs, which matters because throughput tracked host load in the per-request client. We did not run a formal equivalence test, so "consistent" means not distinguishable, not shown equal.

VIII. Results

A. Deployment time and reliability (RQ1)

Table VI reports wall-clock deployment time for the three fast configurations and CI/CD-inclusive deployment time for the fourth. Containerizing a fixed local image adds a consistent ≈0.86s of fixed overhead over a raw process start (1.454s vs. 0.592s mean); adding the full monitoring stack on top of Docker adds a further ≈0.10s (1.552s vs. 1.454s mean) since Compose brings up four containers rather than one but all four start in parallel. The CI/CD-automated path is roughly 35× slower (50.2s vs. 1.454s mean) because it repays the full test-then-build cost on every deployment rather than reusing a pre-built image. This follows from what the path contains (a remote test run and an image build that the other configurations do not perform), so it measures the size of that step, not a property of automation. Of 8 triggered CI/CD runs, 7 completed successfully; the one failure was traced to a transient delay in the workflow being registered by GitHub's

TABLE V
STATISTICAL METHODS, WHY EACH WAS CHOSEN, AND ITS MAIN LIMIT.

Quantity	Method	Why this method	Main limit
Mean of a metric	t-interval, $n-1$ d.f.	Small samples (20 runs per cell, 8 for CI/CD) with unknown variance	Assumes roughly normal means; see the normality check below
Difference between two cases	Welch's t-test with its interval	Variances differ a lot between cases (MTTR std 0.006 s against MTTD std 3.7 s), so equal-variance Student's t would be wrong	Same normality assumption; runs treated as independent
Robustness of that test	Mann–Whitney U, Cliff's δ	Rank-based, no normality assumption, insensitive to outliers; δ is the probability one group exceeds the other minus the reverse	Tests a shift in distribution, not in the mean; ties inflate δ for near-constant metrics
Standardized effect size	Hedges' g with a normal-approximation interval	Puts metrics on one scale; the small-sample correction matters at n=10–20	The interval is approximate and rough at n=10
Success rates (7/8, 20/20)	Wilson score interval	Stays inside $[0, 1]$ and behaves at small n and at 0 or 1, where the Wald interval fails	Describes the binomial rate only
Throughput against CPU	Pearson r with a Fisher-z interval	Measures how strongly run-level throughput tracks host load	Assumes linearity; sequential runs may be autocorrelated
Contention tests (pinning in Exp. 6, background load in Exp. 7)	Difference-in-differences with a percentile bootstrap (10,000 resamples, fixed seed)	A four-group contrast with no closed-form interval; the bootstrap assumes no distribution	Unreliable tails at 10–20 runs per group
Replication verdict	Original mean inside the replication's 95% interval	Fixed before any replication data existed, so it could not be tuned to the result	Ignores the original's own variance and collapses when the replication has almost no variance; not an equivalence test

TABLE VI
DEPLOYMENT TIME BY CONFIGURATION (SECONDS)

Config	Mean	Std	Min	Max	N (fail)
Manual (Baseline A)	0.592	0.020	0.577	0.633	20 (0)
Docker (Baseline B)	1.454	0.107	1.267	1.581	20 (0)
Docker+CI/CD	50.231	3.817	44.404	55.362	8 (1)
Full pipeline	1.552	0.031	1.485	1.636	20 (0)

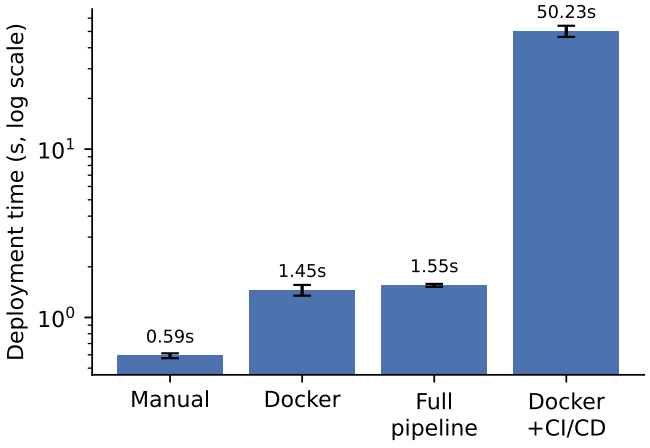


Fig. 4. Deployment time by configuration (log scale; error bars: one SD).

API immediately after being triggered, rather than to a defect in the pipeline itself, and is reported as observed rather than discarded.

Fig. 4 visualizes Table VI on a log scale, since the CI/CD configuration is roughly 30–85× slower than the other three and would otherwise flatten them to indistinguishable bars.

B. Resource utilization, throughput, and monitoring overhead (RQ1, RQ3)

Table VII reports the application container's own resource use and the load test's response characteristics; Table VIII isolates the monitoring stack's own resource draw. Zero errors were observed across all 60 runs (60×300 = 18,000 requests). Point estimates for throughput fall as monitoring is added (2737 → 2641 → 2449 req/s, Docker → +Prometheus → +Grafana); a Welch's t-test (unequal variances, n=20 per case, computed directly from the raw per-run CSV rather than from the summary means) confirms the two endpoints of this trend are statistically distinguishable (Docker vs. +Grafana: t=3.57, p=0.0011, Cohen's d=1.13), but the middle step is not (Docker vs. +Prometheus: t=1.37, p=0.177). This drop did not survive: the replication means have no ordering (4041, 3996 and 4033 req/s), and with a stable client Experiment 6 bounds the monitoring effect at 2.3% (95% interval limit), so we do not attribute the original drop to monitoring (Table XIII, Section VIII-I). Mean and p95 latency both rise numerically but *none* of the three pairwise differences reach significance at α=0.05 (all $p \geq 0.36$); this claim is therefore stated only as a directional observation, not a confirmed effect, and should not be read as strong evidence on its own. Counter-intuitively, the application container's own mean CPU% *falls* over the same progression (11.0% → 3.8% → 1.8%), and here the statistical evidence is strong: Docker vs. +Prometheus is highly

TABLE VII
APPLICATION RESOURCE USE AND RESPONSE CHARACTERISTICS

Config	CPU%	Mem (MB)	p95 (ms)	req/s
Docker	11.02 (3.23)	36.11 (0.41)	5.23 (1.00)	2737 (206)
+Prometheus	3.76 (4.42)	35.68 (0.43)	5.34 (0.80)	2641 (226)
+Grafana	1.79 (3.11)	35.79 (0.96)	5.51 (0.86)	2449 (285)

Mean (std. dev.), $n = 20$ runs of 300 requests each per row. CPU and memory are the application container's; req/s is throughput.

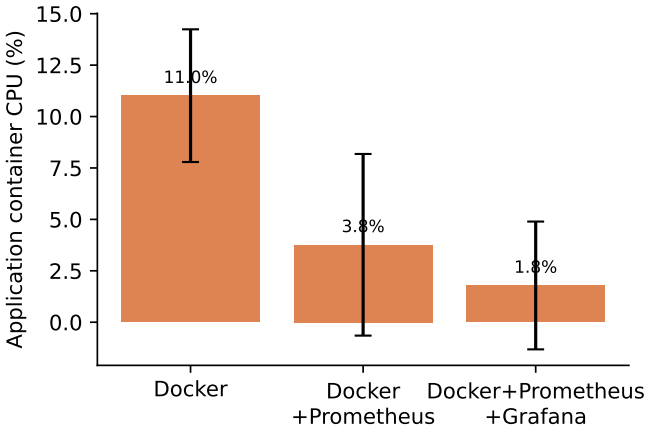


Fig. 5. Application CPU% by monitoring case. The fall is a sampling artifact (Section IX-C).

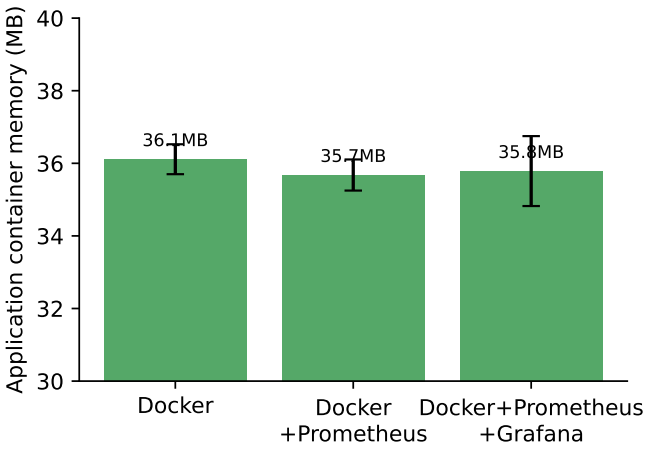


Fig. 6. Application memory by monitoring case (narrow y-axis).

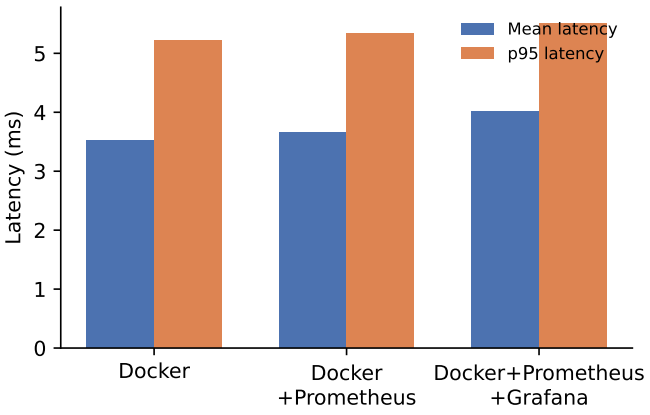


Fig. 7. Mean and p95 latency by monitoring case; no pairwise difference is significant.

significant (t=5.78, p<0.0001, d=1.83, a large effect), though +Prometheus vs. +Grafana is not (p=0.12). We first read this as monitoring containers competing with the application for cores, but that reading is retracted (Section VIII-I): the load window is only ≈0.11s and the `docker stats` reading covers a longer interval, so many runs read as idle (0/20, 12/20 and 16/20 runs below 1% CPU for Docker, +Prometheus and +Grafana) while non-idle runs read ≈8–11% in every case, so the fall in the mean mostly counts idle readings. The `prometheus_client` counters/histograms/gauges add negligible per-request work, as expected.

Table VIII is visualized in Fig. 10, alongside the failure-detection and MTTD/MTTR figures at the end of this section.

C. CPU stress detection (RQ2)

All 20 induced CPU-stress runs were detected by Prometheus, with mean time to detect (MTTD) 5.65s (std 3.68s, min 1.51s, max 18.58s) and mean peak self-reported CPU utilization 96.6% (std 10.0, effectively pegged at one full core, consistent with CPython's GIL limiting a single process to ≈100% of one core regardless of thread count). The detection floor is set by the 2s Prometheus scrape interval combined with the 0.5s poll granularity used to observe it. We did not test what causes the long right tail (up to 18.58s); one candidate is the application's in-process sampling thread being delayed by the stress threads. The signal is the application's own gauge, read through Prometheus, so these are detection times for an application-level signal, and the memory-stress experiment below uses `docker inspect`, not Prometheus.

An initial design used mean `/health` latency as the detection signal instead of process CPU%; it was rejected after finding it barely moved under load, because CPython's GIL round-robins between threads at a short, fixed interval (≈5ms) regardless of how CPU-bound the competing work is, so a trivial handler's queueing delay stays small even under heavy contention.

D. Memory stress and OOM behaviour (RQ2)

All 20 runs triggered a genuine kernel OOM-kill and automatic restart (`RestartCount`=1, confirmed via `docker inspect`), after a mean 5.25 successful allocation requests (std 1.09) and a mean peak reported memory of 135.54 MB (std 0.17) against the 150 MB hard limit. Mean time-to-OOM was 11.58s (std 2.23s); this figure is dominated by the fixed per-iteration overhead of the polling loop itself (a `curl`-equivalent request, a `docker inspect` call, and a `docker stats` call per iteration) rather than by the kernel's actual

TABLE VIII
MONITORING STACK'S OWN RESOURCE DRAW

Container	+Prometheus case		+Grafana case	
	CPU%	Mem (MB)	CPU%	Mem (MB)
cAdvisor	6.13	47.16	6.40	45.39
Prometheus	0.78	59.40	0.80	52.78
Grafana	—	—	0.06	60.70

Mean, $n = 20$ runs per column, from `docker stats`. Kernel counters in Experiment 6 gave 0.027 cores for cAdvisor and 0.005 for Prometheus, about half of these CPU% readings, and the replication read cAdvisor at 2.4%; we do not know which instrument is right for the original host.

OOM response latency, and should not be read as a monitoring detection time in the same sense as Experiments 3, 5, and 6.

E. Why the zero-filled allocation survives: kernel dependence (Experiment 4)

Observation. The unmodified Experiment 4 script held 20 MB blocks of zero-filled memory in a container limited to 150 MB. Originally every run was OOM-killed after about five requests. On the replication kernel none were: 0/20 runs were killed even after 16 requests (≈320 MB held against a 150 MB limit), and the cgroup's `memory.events` showed the limit reached without an OOM kill. Writing one non-zero byte per page (`/memory?fill=true`) restored the original behaviour: 20/20 kills after a mean of 5.0 requests.

Mechanism. `bytearray(n)` writes its zeros, so the memory is resident (an untouched mapping adds about 1 MB; a 200 MB `bytearray` adds 200 MB even with huge pages disabled). With transparent huge pages (THP) set to `always`, those writes are backed by 2 MB pages. Recent kernels can split a huge page that is mostly zero and free its zero subpages (the "underused page" shrinker, enabled here: `shrink_underused=1`). When the cgroup reaches its limit the kernel reclaims memory that way instead of killing the process, so an allocation that is only zeros never exhausts the limit. A non-zero byte in every 4 KiB page makes each huge page non-empty, so nothing can be freed and the OOM killer runs.

Evidence. A standalone probe (`diag_thp.py`; same 150 MB, no-swap limit; up to 20 blocks of 20 MB; 5 runs per cell) crosses the allocation pattern with the THP policy, disabling THP for the probe process alone via `prctl` (Table IX). Only the zero-filled allocation under default THP survives, and it is the only cell where the kernel's underused-huge-page split counter moves (about 170 splits per run). Disabling THP for that one process, with the host, kernel, container and limit unchanged, is enough to restore the OOM kill.

Prediction test. The kernel documentation calls a huge page underused when its zero-filled pages exceed `khugepaged/max_ptes_none` [23], which reads 409 on this host. A 2 MiB page has 512 subpages, so writing one non-zero byte into every k-th 4 KiB page leaves $512 - 512/k$ zero subpages. The rule predicts an OOM kill for $k \leq 4$ (at most 384 zero subpages), survival for $k \geq 6$ (at least 427), and a boundary case at k=5 (410). Table X (`diag_thp_stride.py`, 5 runs per row, default THP) matches: strides 2 and 4 were killed in 5/5 runs with at most one split, the all-zero allocation and strides 6 and 8 survived all 20 blocks in 5/5 runs with 149–171 splits, and the boundary row stopped every run after 12–13 blocks, 4/5 flagged as OOM-killed by Docker (we did not investigate the fifth). This ran on kernel 7.3.0, later than the replication's 7.2.9 whose `max_ptes_none` we did not record, and the 2 MiB alignment of each allocation is not controlled, which blurs the boundary.

TABLE IX
EXPERIMENT 4 DIAGNOSTIC: ALLOCATION PATTERN AGAINST THP POLICY FOR THE PROBE PROCESS (5 RUNS PER ROW; 20 BLOCKS OF 20 MB AT MOST, 150 MB LIMIT). *Blocks* IS THE MEDIAN NUMBER OF BLOCKS REACHED; *THP splits* IS THE RANGE OF THE HOST'S `THP_UNDERUSED_SPLIT_PAGE` COUNTER DELTA PER RUN.

Allocation	THP	OOM-killed	Blocks	THP splits
Zero-filled	default	0/5	20	169–171
Zero-filled	disabled	5/5	7	0
Non-zero	default	5/5	7	0–1
Non-zero	disabled	5/5	7	0

TABLE X
THP STRIDE PROBE: ONE NON-ZERO BYTE IN EVERY k-TH PAGE, 5 RUNS PER ROW, DEFAULT THP, 150 MB LIMIT, `MAX_PTES_NONE` = 409. *Zero pages* IS PER 2 MiB PAGE.

Stride	Zero pages	Split predicted	OOM-killed	Blocks	THP splits
all zero	512	yes	0/5	20	157–171
2	256	no	5/5	7	0–1
4	384	no	5/5	7	0–1
5	410	yes	4/5	13	64–76
6	427	yes	0/5	20	166–170
8	448	yes	0/5	20	149–152

Guest sweep. The host does not let us change these knobs without root, so we repeated the probe in a KVM guest where we are root (Debian 13, kernel 6.12.113, Docker 26.1.5, 4 vCPU, 4 GiB; `vm_thp_matrix.py`, 3 runs per cell). The zero-filled allocation escaped the OOM kill only with THP `always`, `shrink_underused`=1 and `max_ptes_none` below its kernel default of 511 (Table XI); at 511 nothing escaped, shrinker on or off. Values of 0, 200, 409, 500 and 510 all escaped (0/3 killed each, 169–174 splits per run) and 511 did not. The stride boundary matched the host: at 409, strides 2 and 4 were killed in 3/3 runs, stride 5 in 3/3 after 12–13 blocks, and strides 6 and 8 survived all 20 blocks in 3/3.

The host's 409 is not the kernel default. It is written at every boot by the `cachyos-settings` package (`thp-shrinker.conf`), whose own comment gives the default as 511. We then changed it on the host itself, which needs root and was restored afterwards: at 511 the all-zero allocation and strides 4 and 8 were each killed in 5/5 runs with no splits (`diag_thp_host511.csv`), where at 409 the all-zero and stride-8 allocations survive in 5/5 (Table X). The

TABLE XI
GUEST SWEEP, ZERO-FILLED ALLOCATION: OOM-KILLED RUNS OUT OF 3 FOR EACH THP POLICY, `SHRINK_UNDERUSED` SETTING AND `MAX_PTES_NONE` VALUE. ONLY THP `ALWAYS` WITH THE SHRINKER ON AND `MAX_PTES_NONE` =409 ESCAPES.

THP policy	`shrink_underused`	`max_ptes_none` =409	n
always	0	3/3	
always	1	0/3	
madvise	0	3/3	
madvise	1	3/3	
never	0	3/3	
never	1	3/3	

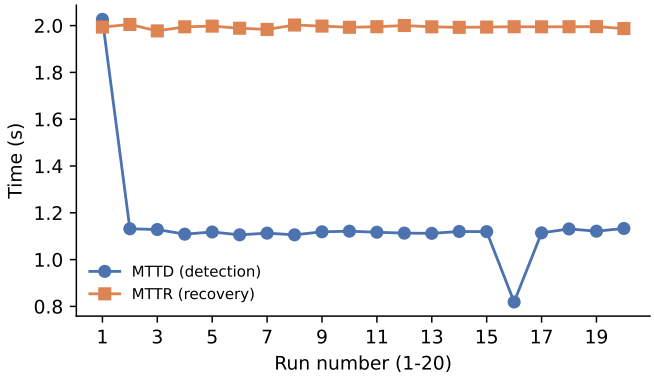


Fig. 8. MTTD and MTTR over the 20 failure-injection runs.

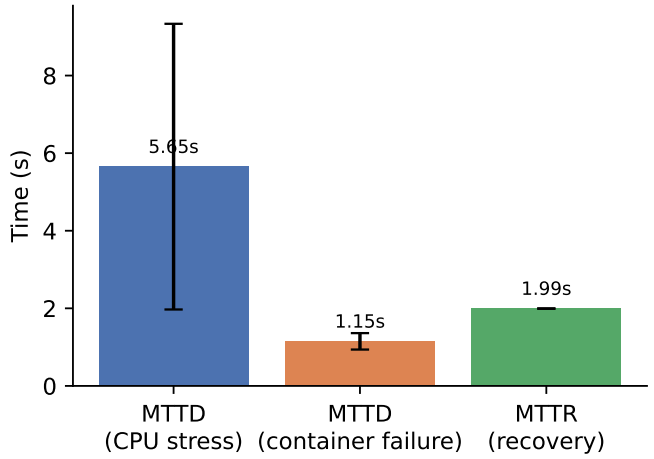


Fig. 9. MTTD (CPU stress), MTTD (killed container) and MTTR, mean ± one SD, original run.

replication installed CachyOS. An installation on the default of 511, as we infer the original Arch one was but did not record, would have left the shrinker inert, which is what the original 20/20 kills look like.

What this means. The OOM-kill result is conditional on the memory being genuinely resident and non-reclaimable. We observed the zero-fill escape on one kernel (7.2.9, THP `always`, `shrink_underused=1`) on one machine; whether other kernels or THP settings behave the same way we did not test. The safe practice does not depend on that answer: a memory-stress test should write non-zero data, and a fault-injection study that depends on memory pressure should report the kernel, THP policy, shrinker setting, `max_ptes_none` and fill pattern. The original finding stands for non-zero memory; for the zero-filled variant it is a property of the kernel configuration.

What is not established. We did not record the original host's THP settings and that installation no longer exists, so we cannot show that they are what differed between the original and replication runs. Either a THP policy of `madvise` or the default `max_ptes_none` of 511 would be consistent with the original result, and we did not test either on the original host. On the host we changed only `max_ptes_none`; the full set of knobs was swept only in the guest, on a different kernel and distribution. The stride probe ran on a later kernel (7.3.0) than the replication (7.2.9), so we have not shown that the replication kernel used the same threshold.

F. Failure detection and recovery (RQ2)

All 20 induced container failures (`docker kill`) were detected by Prometheus's `up{job="app"}` series, mean time to detect (MTTD) 1.15s (std 0.21s, min 0.82s, max 2.03s) — close to the theoretical floor set by the 2s scrape interval combined with 0.3s poll granularity and Docker's own signal-delivery latency. Mean time to recover (MTTR) — container restart to health-check success — was consistently ≈2.0s (std 0.006s, i.e. effectively deterministic), which is essentially Uvicorn's own process startup time inside the container: MTTR here is dominated by application boot, not by any property of the monitoring stack. MTTD is faster and tighter than MTTR in every run, and neither shows a trend over the 20 runs. From this point on, MTTD and MTTR are used throughout in place of "detection time" and "recovery time," consistent with the CPU-stress MTTD reported above and with Figs. 8–9.

G. Monitoring overhead under controlled load (Experiment 6)

Experiment 6 gave two different answers depending on the load client, and the difference is itself a finding. With the Experiment 2 client, which opens a new TCP connection per request, throughput ranged from 92–2,492 req/s and 25 of 120 runs below 500 req/s, with within-case variation of at least 40%. Throughput tracked host load ($r = -0.98$ with host busy cores) and, more tightly, the CPU time of the docker-proxy processes that forward the published port ($r = -0.991$, 95% CI $[-0.994, -0.986]$); docker-proxy used 4.8–9.6 cores in these runs. A separate diagnostic (3 repetitions per client, 40 s each) explains why. The host has 28,232 ephemeral ports; the per-request client reached 42,205 TIME_WAIT sockets, throughput fell from 3,501 req/s in the first 5 s to 120 req/s after 30 s, and docker-proxy spent 9.1 cores of system time against 0.04 of user time (Fig. 11). With keep-alive connections the same host sustained 6,350 req/s throughout with at most 20

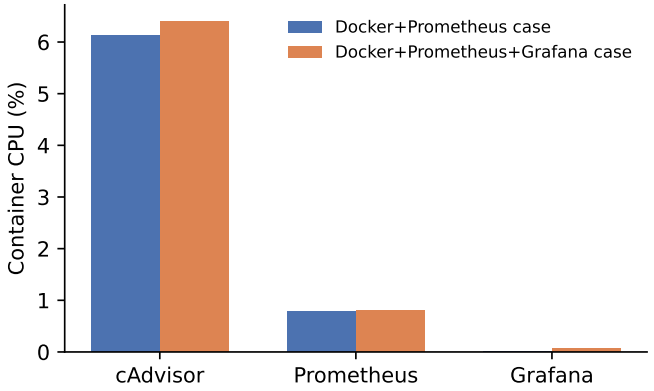


Fig. 10. CPU draw of the monitoring containers (Table VIII), from `docker stats`.

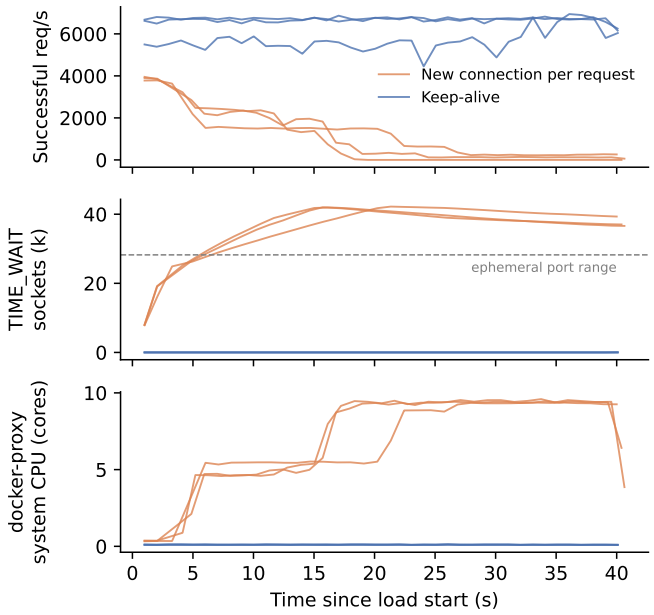


Fig. 11. Per-request client: throughput collapses as TIME_WAIT passes the port range; the keep-alive client is flat (3 repetitions each).

TIME_WAIT sockets and docker-proxy at 0.11 cores. The per-request client therefore mostly measures socket exhaustion on the host over a 20 s load, not the application or the monitoring stack, so its Experiment 6 throughput cannot be attributed to monitoring. Experiment 2's 300-request burst is too short to exhaust the ports, so this does not explain its throughput trend; that trend did not reproduce in the replication (Section VIII-J), and the keep-alive result below bounds the effect at 2.3% (95% interval limit).

The keep-alive client is stable (within-case variation at most 2.3%) and is the one we use to bound monitoring overhead. It is not the bottleneck: the application container used 1.03–1.04 cores, a saturated single Uvicorn process, the load generator about 0.54 cores, and four load-generator processes raised Docker-only throughput by only about 3.5% (6,788 against

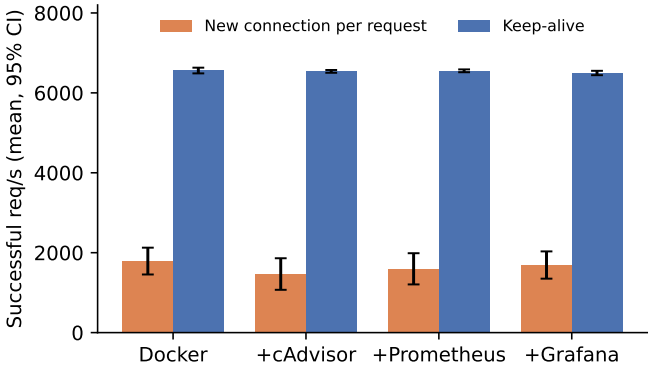


Fig. 12. Experiment 6 throughput per case, both clients (mean, 95% CI, unpinned, one load generator).

6,559 req/s). The bound therefore applies to a closed-loop, server-saturated configuration; we did not run an open-loop fixed-rate test. Against Docker alone, the full stack (+Grafana) differed by -62 req/s, 95% CI $[-149, +25]$ (-0.9%), Hedges' $g = -0.45$ $[-1.07, +0.18]$, and no monitored case moved throughput by more than 0.9% (Fig. 12). The monitoring containers used 0.032 cores for cAdvisor, Prometheus and Grafana together, with the application container at 1.03–1.04 cores. The CPU-pinned arm gave a difference-in-differences $+8$ req/s, 95% CI $[-94, +105]$, so it neither supports nor excludes a contention effect of that size. One configuration does detect a difference: against Docker alone, +Grafana with four load-generator processes (-1.4%, 95% CI $[-159, -36]$ req/s) is lower, and the interval excludes zero (10 Docker-only and 9 +Grafana runs; one +Grafana run is missing from the log). We did not correct for the number of contrasts examined, and the 20-run and 10-run designs were not powered for effects this small. We therefore report a bound, not an absence: on this host, with a client that does not exhaust ports, the largest observed loss is 1.4% of throughput and no 95% interval in the unpinned contrasts reaches beyond a 2.3% loss. One of the two keep-alive configurations detects a loss of the observed size. The runs came in randomized blocks, so we also paired each case with the Docker-only run of the same block; for that configuration the paired difference was -104 req/s, 95% CI $[-160, -48]$ (9 blocks), no tighter than the independent-groups interval, so block-to-block drift is not what limits the precision. We did not identify the mechanism.

H. Monitoring cost under background CPU load (Experiment 7)

Contention predicts that the monitoring cost grows as the host gets busier. Experiment 7 repeated the keep-alive measurement for Docker alone and the full stack while a duty-cycle burner on all 16 logical CPUs held a background target of 0, 20, 40 or 60% (cells shuffled within blocks, 15 runs per cell, burner CPU read from `/proc`; achieved 2.4 busy cores, 5.5 busy cores, 8.6 busy cores and 12.0 busy cores). The monitoring containers used 0.038–0.053 cores, and the

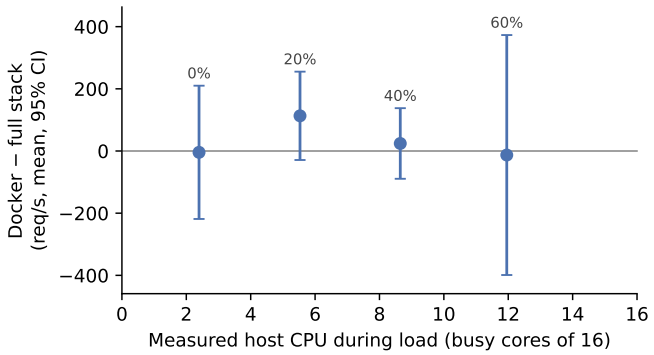


Fig. 13. Experiment 7: monitoring penalty against measured host CPU; all intervals include zero.

penalty (Docker-only minus full-stack throughput) did not grow (Fig. 13): -4 req/s, 95% CI $[-219, +210]$ (-0.1%) at 0%; $+113$ req/s, 95% CI $[-29, +255]$ ($+2.6\%$) at 20%; $+24$ req/s, 95% CI $[-89, +138]$ ($+0.6\%$) at 40%; -13 req/s, 95% CI $[-399, +373]$ (-0.4%) at 60%. The test fixed before data collection, a difference-in-differences -9 req/s, 95% CI $[-424, +389]$ between 60% and 0%, leaves contention not supported. With run-to-run variation of 3–6%, rising to up to 16% at the 60% level, the experiment only rules out a load-dependent penalty above about 11% of throughput, so it confirms that there is no large contention effect and adds nothing finer than Experiment 6.

I. Revised explanation of the CPU-contention finding

Original claim. In the original run the application's CPU reading fell (11.0% $\rightarrow$ 3.8% $\rightarrow$ 1.8%) and throughput fell (2737 $\rightarrow$ 2641 $\rightarrow$ 2449 req/s, -10.5%) as monitoring containers were added. We explained both by host-level CPU contention: the monitoring stack competes with the application for cores, so the application is scheduled less. We no longer hold this as a supported conclusion.

An alternative the original design cannot exclude. Experiment 2 ran its cases in a fixed order on one stack instance each, so the monotone fall in throughput is also what a drift over the session (thermal state, background activity) would produce. The raw file holds two further passes of the same script on the same installation, not used for any table. The second ran straight after other experiments: 14 of its 40 monitored runs had request errors, and the means were 2895, 2597 and 1799 req/s. The third ran alone, with no errors, and gave 1660, 1648 and 1595 req/s (-0.7% and -3.9% against Docker), a Docker-only level 40% below the reported pass on that same installation. The reported pass is the first of three, kept because it was the pre-planned one, not because it was typical. Across passes the ordering Docker $>$ +Prometheus $>$ +Grafana recurs, but its size ranges from -3.9% to -10.5% in the clean passes, and Experiment 2 cannot separate monitoring from session position.

Interleaved re-run (Experiment 2i). To separate the two we repeated Experiment 2's load with the three cases plus a second, identical Docker-only arm (an A/A arm), a fresh stack for every run, and the four cases in shuffled order within each of 20 blocks (80 runs, 10.6 min). It ran on a third software state (kernel 7.3.0, Docker 29.9.2, balanced power profile on AC power), so only its within-session contrasts are interpretable. Throughput was 2,707, 2,629 (A/A), 2,689 and 2,662 req/s for Docker, A/A, +Prometheus and +Grafana, with no request errors. Against Docker alone, the A/A arm differed by -78 req/s, 95% CI $[-254, +99]$ (-2.9%), +Prometheus by -18 req/s, 95% CI $[-175, +139]$ (-0.7%) and +Grafana by -45 req/s, 95% CI $[-219, +128]$ (-1.7%); against both Docker-only arms pooled, +Grafana differed by -7 req/s, 95% CI $[-169, +156]$. The identical-configuration arm moves as far as either monitored case, so no monitoring effect is distinguishable from this design's noise floor. Throughput drifted down across the session by -21.8 req/s per block, 95% CI $[-31.2, -12.4]$, about 16% over the 20 blocks, with no effect of position inside a block ($+33$ req/s per position, 95% CI $[-21, +87]$). Assigning the same runs to a fixed order after the fact (Docker and A/A from the first seven blocks, +Prometheus from the next seven, +Grafana from the last six; an illustration chosen post hoc, not a test) gives 2,762, 2,728 and 2,383 req/s, a -13.7% fall to the full stack, the size of the original -10.5%. The original drop is therefore what a fixed case order produces under session drift. We cannot show that the original session drifted by the same amount.

What contention predicts, and what Experiment 6 found. Contention was also a weak hypothesis on priors: the application is a single Uvicorn process that uses about one core on a host with 16 hardware threads, and the monitoring stack uses about 0.03 cores.

1) *The monitoring containers should use meaningful CPU.* They did not: about 0.03 cores in total, measured from kernel counters.
2) *Throughput should fall when monitoring is added, with a stable client.* It fell by at most 0.9% with one load-generator process and by 1.4% with four, against the 10.5% originally read.
3) *Moving the monitoring containers to cores the application does not use should remove the drop.* The pinning test is inconclusive in both clients: a difference-in-differences of $+376$ req/s, 95% CI $[-343, +1061]$ with the per-request client and $+8$ req/s, 95% CI $[-94, +105]$ with the keep-alive client. It neither supports contention nor excludes an effect of that size.
4) *Runs with more monitoring CPU should have lower throughput.* With the stable client they do not: across all runs $r = -0.13$, 95% CI $[-0.34, +0.09]$, and within a case $r = -0.07$, 95% CI $[-0.28, +0.16]$. With four load-generator processes the pooled association is $r = -0.63$, 95% CI $[-0.84, -0.25]$, but within a case it is $r = -0.16$, 95% CI $[-0.58, +0.31]$: it comes from the Docker-only against monitored contrast, not from the amount of monitoring CPU.
5) *The monitoring penalty should grow as the host gets*

busier. It did not (Experiment 7, Section VIII-H): no level from 0 to 60% background load shows a penalty whose interval excludes zero, and there is no trend across levels.

Revised explanation. (i) The fall in application CPU% is a measurement artifact of a 0.11 s load window sampled by a longer interval (Section VIII), not a scheduling effect. (ii) The original throughput drop cannot be attributed to monitoring. It did not reproduce (replication means of 4041, 3996 and 4033 req/s have no ordering), and with a client that does not exhaust ports the monitoring effect is at most 2.3% (95% interval limit), several times smaller. What did drive throughput in the per-request client was socket exhaustion and host load (Section VIII-G). (iii) What remains is a loss of about 1.4% in one of two keep-alive configurations, associated with about 0.03 cores of monitoring CPU. We did not identify its mechanism; contention is one candidate that we have not tested and that the evidence above does not favour over run-to-run variation; Experiment 7 found no growth with load, though it can detect only large growth.

Status. Retracted as a supported conclusion. Experiments 6 and 7 exclude large contention effects on this host, including a large load-dependent one; they do not exclude effects of about 1% or smaller, and neither the pinning test nor the load test is precise enough to say whether any such effect is contention.

J. Same-hardware replication

Experiments 1–5 were repeated with the unmodified scripts on the same hardware on a later date (20 runs per configuration), but on a freshly installed operating system (CachyOS, kernel 7.2.9, Docker 29.8.2) in place of the Arch/Omarchy 7.1.9 installation of Table III, so this tests robustness to the software environment rather than to run-to-run noise alone. Results were written to a separate directory so the data behind Tables VI, VII and VIII is untouched. Table XII gives the original and replication means with 95% confidence intervals (`validate.py`); a metric is marked *consistent* only if the original mean lies inside the replication's interval. Most absolute values moved well outside the original intervals, while the relative structure that the research questions rely on held for deployment ordering (including the Docker+CI/CD arm, 8 real GitHub Actions runs, which replicated within its original interval), MTTR and peak CPU under stress (Section XI lists what did not reproduce and why). The original run remains the paper's reported dataset because it was the pre-planned measurement and every table is derived from it (for Experiment 2 it is the first of three passes on that installation, Section VIII-I); the replication is reported as a robustness check, not as a replacement.

IX. Four Measurement Pitfalls

Each pitfall below turned a property of the measurement apparatus into an apparent property of the system under test, and none was visible in the summary statistics of a single campaign. P1 and P4 surfaced only because the study was repeated in a second software environment; P2 and P3 only because the same quantity was re-measured with a different instrument. Table I summarizes them.

A. P1: Zero pages are not memory pressure

Symptom. The memory-stress test that OOM-killed the container in 20/20 original runs killed it in 0/20 replication runs, although 16 requests (≈320 MB) succeeded against a 150 MB limit. *Mechanism (predicted and tested, Table X).* The allocation is zero-filled. On the replication kernel, with transparent huge pages enabled and the underused-huge-page shrinker active (the kernel documentation calls a huge page underused when its zero-filled pages exceed `max_ptes_none` [23]), the kernel splits mostly-zero huge pages and frees their zero subpages under cgroup pressure, so the limit is reached without an OOM kill. The shrinker is inert at the kernel default `max_ptes_none`=511, so the escape depends on a distribution setting (Table XI). *Evidence.* Disabling THP for the one probe process restores the kill, and only the surviving cell moves the split counter (Table IX, Section VIII-E), and a stride probe matches the threshold in the kernel documentation (Table X). *Fix.* Before interpreting a stress result, confirm from the kernel that the injected fault occurred: check `memory.events` for `oom_kill` [24], write non-zero data to every page, and report the kernel, THP policy and fill pattern.

B. P2: The load client measures the socket layer

Symptom. With a client that opens a new TCP connection per request, throughput varied three-fold between runs and tracked the CPU time of `docker-proxy`. *Mechanism.* The client exhausts the host's ephemeral ports within seconds; TIME_WAIT sockets pile up and `docker-proxy` spends its time in the kernel, so throughput measures the host's socket layer, not the application. *Evidence.* Per-second TIME_WAIT and proxy CPU series show the collapse, and a keep-alive client on the same host is flat (Fig. 11, Section VIII-G). *Fix.* Use keep-alive connections, record TIME_WAIT and client and proxy CPU, and check within-case variation before comparing cases. Because the Docker daemon runs a userland proxy for loopback traffic by default and every request here crosses it [25], disabling it or loading from inside the container network would also avoid the problem; we tested neither. The TIME_WAIT state that fills up keeps a closed connection's address pair reserved for a fixed period [26].

C. P3: The sampling window outlasts the load

Symptom. The application's CPU reading fell as monitoring containers were added, which we first reported as CPU contention. *Mechanism.* The Experiment 2 load lasts about 0.11 s and a `docker stats` reading covers a longer interval, so many runs read idle and the mean counts how often the sample missed the burst. *Evidence.* The per-run readings are bimodal, and kernel counters read over the exact load window show the monitoring containers using about 0.03 cores with no contention effect (Sections VIII and VIII-I). *Fix.* Use a fixed-duration load and read cgroup `cpu.stat` at the edges of the load window.

TABLE XII

ORIGINAL VS. SAME-HARDWARE REPLICATION ON A REINSTALLED OS (20 RUNS EACH). INTERVALS ARE t-BASED 95% CIS ON THE MEAN (SAMPLE SD); Δ% IS REPLICATION RELATIVE TO ORIGINAL. THE EXPERIMENT 4 ROWS ARE FROM THE UNMODIFIED ZERO-FILLED SCRIPT (0/20 OOM KILLS, SEE SECTION XI); THE NON-ZERO-FILL ROWS COME FROM THE VARIANT THAT KILLS THE CONTAINER. A ROW IS *consistent* ONLY IF THE ORIGINAL MEAN LIES INSIDE THE REPLICATION'S INTERVAL. THIS RULE IS STRICT WHEN THE REPLICATION RUNS ARE NEARLY IDENTICAL (5 REQUESTS AND 11.06 S IN EVERY NON-ZERO-FILL RUN), SO THOSE ROWS READ *different* ALTHOUGH THE VALUES ARE WITHIN ONE REQUEST, 1 MB AND 0.5 S OF THE ORIGINAL. THE CPU% ROWS MARKED CONSISTENT REST ON THE BIMODAL MEASUREMENT DISCUSSED IN SECTION XI. EXPERIMENT 6 HAS NO ORIGINAL RUN AND IS NOT IN THIS TABLE. GENERATED FROM `RESULTS/ORIGINAL_VS_REPLICATION.CSV`.

Metric	Original mean [95% CI]	Replication mean [95% CI]	Δ%	Verdict
Deploy time, manual (s)	0.59 [0.58, 0.60]	0.42 [0.41, 0.43]	-28.9	different
Deploy time, Docker (s)	1.45 [1.40, 1.50]	1.05 [1.04, 1.06]	-27.7	different
Deploy time, full pipeline (s)	1.55 [1.54, 1.57]	1.37 [1.31, 1.43]	-11.8	different
Deploy time, Docker+CI/CD (s)	50.23 [46.42, 54.04]	48.60 [44.05, 53.15]	-3.2	consistent
Throughput, Docker (req/s)	2737.07 [2638.29, 2835.84]	4040.66 [3984.28, 4097.04]	$+47.6$	different
Throughput, +Prometheus (req/s)	2640.65 [2532.08, 2749.23]	3995.78 [3944.85, 4046.70]	$+51.3$	different
Throughput, +Grafana (req/s)	2448.85 [2311.85, 2585.84]	4033.22 [3995.02, 4071.43]	$+64.7$	different
p95 latency, Docker (ms)	5.23 [4.75, 5.70]	3.16 [2.96, 3.36]	-39.5	different
p95 latency, +Prometheus (ms)	5.34 [4.95, 5.72]	3.14 [3.03, 3.25]	-41.2	different
p95 latency, +Grafana (ms)	5.51 [5.09, 5.92]	3.03 [2.97, 3.08]	-45.1	different
App CPU%, Docker	11.02 [9.47, 12.56]	7.42 [6.37, 8.46]	-32.7	different
App CPU%, +Prometheus	3.76 [1.64, 5.88]	3.50 [1.81, 5.18]	-7.1	consistent
App CPU%, +Grafana	1.79 [0.29, 3.28]	1.91 [0.65, 3.16]	$+6.6$	consistent
App mem, Docker (MB)	36.11 [35.91, 36.31]	37.66 [37.49, 37.82]	$+4.3$	different
App mem, +Prometheus (MB)	35.68 [35.47, 35.88]	37.74 [37.50, 37.98]	$+5.8$	different
App mem, +Grafana (MB)	35.79 [35.32, 36.25]	37.72 [37.49, 37.95]	$+5.4$	different
cAdvisor CPU%, +Prometheus	6.12 [4.58, 7.67]	2.41 [1.80, 3.02]	-60.7	different
cAdvisor CPU%, +Grafana	6.40 [4.98, 7.82]	2.19 [1.60, 2.78]	-65.8	different
Prometheus CPU%, +Prometheus	0.78 [0.40, 1.16]	0.62 [0.26, 0.98]	-20.4	consistent
Prometheus CPU%, +Grafana	0.80 [0.47, 1.13]	0.51 [0.18, 0.84]	-36.1	consistent
Grafana CPU%, +Grafana	0.06 [0.01, 0.11]	0.05 [0.03, 0.07]	-22.5	consistent
CPU-stress MTTD (s)	5.65 [3.88, 7.42]	2.41 [2.07, 2.75]	-57.3	different
CPU-stress peak CPU%	96.59 [91.79, 101.39]	95.13 [89.69, 100.57]	-1.5	consistent
Time to OOM (s)	—	—	—	no values (0/20)
Peak memory (MB)	135.54 [135.46, 135.62]	138.59 [137.53, 139.65]	$+2.3$	different
Requests before OOM	5.25 [4.73, 5.77]	16.00 [16.00, 16.00]	$+204.8$	different
Failure MTTD (s)	1.15 [1.05, 1.25]	3.14 [3.07, 3.20]	$+172.9$	different
MTTR (s)	1.99 [1.99, 2.00]	1.98 [1.94, 2.02]	-0.7	consistent
Time to OOM, non-zero fill (s)	11.58 [10.51, 12.66]	11.06 [11.06, 11.06]	-4.5	different
Peak memory, non-zero fill (MB)	135.54 [135.46, 135.62]	136.41 [136.25, 136.56]	$+0.6$	different
Requests before OOM, non-zero fill	5.25 [4.73, 5.77]	5.00 [5.00, 5.00]	-4.8	different

D. P4: Absolute numbers do not travel

Symptom. After the operating-system reinstall, deployment times, throughput and detection latencies moved by 12–173% on the same hardware (Section VIII-J). The reinstall is not the only source: on the original installation, an isolated re-run of Experiment 2 gave a Docker-only throughput of 1660 req/s against 2737 req/s in the reported pass, and within a single 10.6-minute session the interleaved re-run drifted by about 16% (Section VIII-I). Failure detection moved from 1.15 to 3.14 s in the replication, and two isolated Experiment 5 sessions on the current software state gave 3.10 s (SD 0.18 s) and 3.06 s (SD 0.15 s), so that shift is stable within an environment rather than session noise; we did not identify its cause. Later, the same keep-alive Docker-only measurement gave 6,559 req/s in Experiment 6 and 4,759 req/s in Experiment 7, a week apart (-27%), with unattributed background CPU of 0.63 cores against 0.32 cores in Experiment 6. *Mechanism, partly tested.* The campaign snapshots show that Experiment 7 ran on battery with the balanced power profile, while Experiment 6 ran on AC with the performance profile. We tested that directly on AC, alternating profiles in four blocks: medians of 5,035 req/s under performance and 4,889 req/s under balanced, 12 runs per profile. Across all runs the difference is -278 req/s, 95% CI $[-945, +390]$, because 2 runs at half speed fell in a performance block; without them (a post-hoc exclusion) it is $+171$ req/s, 95% CI $[114, 228]$ ($+3.5\%$). The power profile therefore explains a few percent, not the gap. Between the two experiments Docker was updated from 29.8.2 to 29.9.0 and the kernel package was rebuilt (7.2.9-1 to 7.2.9-2) by routine system updates; `docker-proxy` carries every request, so these are the remaining candidates, but we did not test them. *Fix.* Record the full software environment, including package versions and power state, with every run; pin versions for the duration of a study; randomize the order of configurations within a session; and run one configuration against itself in a separate session (an A/A comparison) to measure the noise floor before reading any difference between configurations.

X. DISCUSSION

A. Answers to the case-study questions

RQ1 (containerization's effect on deployment time and resource use). Containerization has a small, consistent fixed cost at deployment time ($\approx$0.86s here) and a resource footprint dominated by the application's own working set (36 MB) rather than by container overhead per se — consistent with the classic finding in Felter et al. [1] that containers add near-zero CPU/memory overhead relative to native execution. A

TABLE XIII

ORIGINAL AGAINST REPLICATION: WHICH FINDINGS HELD, COMPUTED FROM TABLE XII, THE RAW DATA AND THE EXPERIMENT 6 AND 7 RESULTS (`REPRODUCIBILITY_TABLE.PY`). *Reproduced*: EVERY METRIC IN THE ROW HAS THE ORIGINAL MEAN INSIDE THE REPLICATION'S 95% INTERVAL, OR THE STATED QUALITATIVE RESULT HELD. *Environment-sensitive*: EVERY METRIC LIES OUTSIDE IT, OR THE QUALITATIVE RESULT CHANGED. *Not supported by replication*: A FIRST-CAMPAIGN EXPLANATION THAT THE REPLICATION AND EXPERIMENTS 6–7 CONTRADICT. *Inconclusive*: THE DATA CANNOT DECIDE. WHERE THE REPLICATION INTERVAL IS WIDE (THE CPU ROWS), "REPRODUCED" MEANS NOT DISTINGUISHABLE, NOT SHOWN EQUAL.

Finding	Evidence (change is replication relative to original)
Reproduced	
Deployment ordering (manual $<$ Docker $<$ full pipeline) and CI/CD deployment time	Order preserved; CI/CD time -3.2%
Recovery time (MTTR) after a container kill	Change -0.7%
Peak CPU under CPU stress	Change -1.5%
Prometheus and Grafana CPU draw	Change -36.1 to -20.4%
OOM kill when the allocated memory is non-zero	20/20 killed, within 1 request and 1 MB of the original
Environment-sensitive	
Absolute deployment time, throughput, latency and memory	deployment -28.9 to -11.8%; throughput $+47.6$ to $+64.7\%$; p95 -45.1 to -39.5%; memory $+4.3$ to $+5.8\%$
Detection times	CPU stress -57.3%; failure $+172.9\%$
cAdvisor CPU draw	Change -65.8 to -60.7%
OOM kill when the allocated memory is zero-filled	20/20 killed originally, 0/20 in the replication
Not supported by replication	
Monitoring lowers throughput through host CPU contention	Replication means 4041, 3996, 4033 req/s, not monotone; monitoring used 0.03 cores, the largest keep-alive loss was 1.4% and no 95% interval reached beyond 2.3%; the penalty did not grow with background load (Exp. 7)
Application CPU% falls because monitoring competes for CPU	The reading falls again ($7.4 \rightarrow 3.5 \rightarrow 1.9\%$) but is bimodal: a 0.11 s load sampled over a longer window
Inconclusive	
A small throughput loss from monitoring	Detected in 1 of 2 keep-alive configurations (-1.4%); mechanism not identified
CPU pinning removes the monitoring penalty	Difference-in-differences $+376$ req/s, 95% CI $[-343, +1061]$, from the unreliable per-request client
CI/CD deployment success rate	7/8 [0.53, 0.98] originally, 8/8 [0.68, 1.00]; too few runs to estimate the rate
THP policy is what changed the zero-fill OOM result	Sufficient on the replication host (Table IX); the original THP setting was not recorded
Cause of the 27% Exp. 6 to Exp. 7 throughput gap	Power profile explains a few percent (Section IX-D); Docker and kernel updates untested

CI/CD-automated deployment is roughly $35\times$ slower end-to-end because it includes the build and test run; that is true by construction and we report it only so the size of the step is on record, not as a finding about automation. The automated path completed 7 of 8 runs; the 95% Wilson score interval for 7 successes out of 8 trials is $[52.9\%, 97.8\%]$, so the data are consistent with anything from roughly half to nearly all deployments succeeding. The study has no manual-deployment arm and cannot say whether automation improves reliability.

RQ2 (monitoring's effect on detecting degradation and failure). Prometheus detected every induced CPU saturation event and every container failure, 20/20 in each experiment in both the original run and the replication. That detection succeeds is the finding that holds. How fast it succeeds did not: the original run detected failures (mean 1.15s) about five times faster than CPU saturation (mean 5.65s), but the replication inverted the ordering (3.14s against 2.41s), so both detection times are environment-sensitive (Table XIII) and we draw no conclusion about which kind of event is detected faster. Unlike the pass/fail scenarios of Ekanayaka et al. [4], these are repeated timed measurements, but on one host they bound only this configuration.

RQ3 (monitoring overhead). The original run suggested a structurally unusual overhead: monitoring containers competing with the application for cores, cutting throughput by 10.5% while the application's own CPU% fell. Experiment 6 retracts that explanation (Section VIII-I). The CPU% fall was a measurement artifact; the monitoring containers drew about 0.03 cores; and with a stable client throughput was within 1.4% of Docker-only, a loss one of two configurations detects and whose mechanism is unidentified. Under background CPU load up to 60% the penalty did not grow (Experiment 7). Our answer to RQ3 is therefore a bound: on this host the Prometheus–Grafana stack costs at most about 2.3% of throughput (95% interval limit; 1.4% at worst observed) and about 0.03 cores, and cAdvisor dominates the tooling's own CPU draw in the original run but not in the replication (6.1% against 2.4%).

RQ4 (what survives a change of environment). Table XIII gives the full classification. Relative and structural findings held: the deployment ordering, the CI/CD build-and-test cost, recovery time, peak CPU under stress, the Prometheus and Grafana CPU draw, and the OOM kill when memory is

resident. Absolute timings, throughput, latency and detection times did not, and two explanations offered in the first campaign (CPU contention as the cause of the throughput drop and of the falling CPU reading) are not supported. Two of the held findings are weaker evidence than the others: the CI/CD build-and-test time runs on GitHub-hosted runners that the local reinstall does not touch, and the deployment ordering follows from what each configuration starts. The findings that failed to reproduce trace to P1–P3 where we could test the mechanism; for the rest (absolute timings, detection times) the reinstall changed the operating system, kernel, Docker version and an unrecorded power state together, so we cannot say which change mattered.

XI. Limitations

- **Replication on the same hardware did not reproduce most absolute numbers.** Re-running Experiments 1–5 with the unmodified scripts on the same machine after the operating system was reinstalled (CachyOS, kernel 7.2.9, Docker 29.8.2, replacing Arch/Omarchy 7.1.9; performance power profile; 20 runs each) (Table XII) gave deployment times 12–29% faster, throughput about 48% higher, CPU-stress detection of 2.41s rather than 5.65s and failure detection of 3.14s rather than 1.15s, with the mean outside the original 95% interval for most metrics. Only the Docker+CI/CD deployment time (48.6 s vs. 50.2 s; 8/8 real GitHub Actions runs succeeded vs. 7/8 originally), MTTR, peak CPU under stress, the Prometheus/Grafana CPU figures and the application CPU% in the +Prometheus and +Grafana cases stayed consistent; the replication success rate has a 95% Wilson interval of [0.68, 1.00], so it does not contradict the original 7/8. The deployment ordering (manual, then Docker, then full pipeline) was preserved, but the throughput ordering across monitoring cases was not (4041, 3996 and 4033 req/s for Docker, +Prometheus and +Grafana); absolute values are specific to one software environment.
- **The CPU-contention explanation is not supported.** Experiment 2's application CPU% is bimodal because its ≈0.11s load window is sampled by a longer `docker stats` interval, so the reported fall with monitoring is mostly runs that read as idle. A new Experiment 6 (20 runs per case, fixed 20s load, CPU from kernel counters over exactly the load window, randomized case order, optional CPU-pinned arm) found the monitoring containers used ≈0.03 cores in total, and the per-request client showed no throughput difference from Docker-only whose 95% interval excluded zero. That client is unreliable: its throughput varied roughly three-fold, tracked host load ($r = -0.98$ with host busy cores) and collapsed as TIME_WAIT sockets exhausted the ephemeral ports (Section VIII-G), so its pinning result (difference-in-differences $+376$ req/s, 95% CI $[-343, +1061]$) is inconclusive. A keep-alive client is stable and bounds the monitoring cost at 2.3% of throughput (95% interval limit), but one of its two configurations (four load-generator processes, 9–10 runs per case, no multiple-comparison correction) detects a loss of that size whose mechanism we did not identify. The experiments can rule out large monitoring overhead on this host; they cannot rule out small effects.
- **What is and is not reproducible.** From the committed raw data, `make reproduce` rebuilds every table, statistic and figure in the paper without Docker: unit tests, summary tables checked against the committed copies, every number quoted for Experiments 1–5 and 6 found verbatim in the paper, confidence intervals and effect sizes, and every plotted figure compared pixel for pixel. Two things are outside that guarantee. First, re-running the experiments (`make campaign`) collects new measurements, and the replication shows these differ with the software environment, so a rerun reproduces the procedure, not the numbers. Second, Fig. 3 is a screenshot of a live dashboard and differs on every capture; only the underlying Prometheus series are saved. The original 2026-09 host (kernel, THP setting) no longer exists, so the original runs themselves cannot be re-executed under their original conditions.
- **Experiment 4 depends on kernel behaviour** (mechanism and evidence in Section VIII-E). The retained allocation is zero-filled. On the replication kernel, which enables transparent hugepage shrinking of zero-filled pages (`shrink_underused=1`), the kernel reclaimed that memory under cgroup pressure, so 0/20 containers were OOM-killed even though 16 requests (≈320 MB) succeeded against a 150 MB limit. With the allocation writing a non-zero byte to every page (`/memory?fill=true`, default off), 20/20 runs were OOM-killed after a mean 5.0 requests with peak memory 136.4 MB, close to the original result (5.25 requests, 135.5 MB; the interval rule of Table XII marks these rows different because the replication runs are nearly identical). The OOM finding therefore holds when the memory is genuinely resident; the original zero-filled variant is kernel dependent. The original run was on a different distribution and kernel (Arch 7.1.9 vs. CachyOS 7.2.9); the replication host has transparent hugepages set to `always`. We did not record the original host's THP setting and the original installation no longer exists, so we cannot confirm that this setting is what differed; the distribution default being `madvise` is a plausible but untested explanation.
- **cAdvisor produced no usable per-container series.** This host's Docker stores images through the containerd snapshotter (`docker info`: storage driver `overlayfs`, `io.containerd.snapshotter.v1`), not the classic `overlay2` layout. cAdvisor v0.49.1 logged that it could not identify the read-write layer of any container and attached no container name or image label, so Prometheus queries filtered by container returned nothing. Three consequences follow. The

detection experiments use the application's own Prometheus metrics, so their detection times are for an application-level signal, not for container-level CPU. The per-container CPU and memory figures come from `docker stats` and cgroup counters, not from the monitoring stack. cAdvisor still ran and was scraped in every monitored case, so its CPU is included in the overhead figures, but we did not measure the cost of a cAdvisor that resolves labels. We tested one cAdvisor version with one storage driver; this is not a statement about cAdvisor in general, and other versions or the `overlay2` driver may behave differently.

- **GHCR authentication.** The Docker+CI/CD deployment step could not pull from GitHub Container Registry (the CLI token used lacks the `read:packages` scope, which would require an interactive browser re-authentication out of scope for an unattended experiment run) and instead deploys the locally built image from the same commit and Dockerfile CI just built. The CI timing itself (test + build + push) is unaffected and fully real.
- **Reduced n for the CI/CD arm.** Only 8 (vs. 20 for the other three deployment configurations) real GitHub Actions runs were triggered, because each run costs ≈45–65s of real CI wall time and consumes Actions minutes; this narrows the confidence interval on the CI/CD reliability and timing figures relative to the other three configurations.
- **Single-host measurement.** All experiments ran on one physical machine with no isolation from other host activity beyond what Docker itself provides; absolute timing figures (especially the sub-2s deployment times) should be read as relative comparisons between configurations on this host, not as portable absolute benchmarks.
- **Experiment 7 precision and conditions.** Its 15 runs per cell bound only a large load-dependent penalty. It ran on battery with the balanced power profile (Section IX-D); that state was constant within the experiment, so its within-experiment contrasts stand, but its throughput levels are not comparable with Experiment 6. The 60% target loads 16 logical CPUs on 8 physical cores, so it is not saturation, and SMT sharing adds noise at that level. Its background-CPU precondition was relaxed from 0.5 to 1.5 cores because the interactive desktop could not be closed; unattributed CPU is recorded per run.
- **Detection thresholds are fixed, not tuned.** The 60% CPU-percent threshold (Experiment 3) and the memory container limit (150 MB, Experiment 4) were chosen to make the induced condition reliably observable within a short polling window rather than to model a specific production SLO; detection-time figures should be read relative to these thresholds, not as threshold-independent constants.
- **Not every reported difference is statistically significant.** Welch's t-tests on the raw per-run data (`statistical_analysis.py`) confirm the Docker-to-+Grafana endpoints of the throughput and CPU% trends in Section VIII ($p<0.01$ for both), but two things reported only as directional observations do not reach significance at $\alpha=0.05$: the latency increase across all three monitoring cases ($p \geq 0.36$ for every pair), and the middle step of the throughput trend (Docker vs. +Prometheus, $p=0.177$). Means and standard deviations alone, without this check, would have overstated how uniformly these effects hold run-to-run.

XII. Conclusion

A small, carefully logged study of a containerized CI/CD pipeline with Prometheus–Grafana monitoring produced a clean set of findings on its first run. Repeating it on the same hardware after an operating-system reinstall showed which of them were properties of the pipeline and which were properties of how we measured it. What held: the ordering of deployment times, the cost of re-running build and test on every CI/CD deployment, recovery time after a container kill, and an OOM kill when memory is resident. Monitoring cost at most about 2.3% of throughput (95% interval limit) and about 0.03 cores with a stable client, and the first campaign's CPU-contention explanation is withdrawn. Whether a loss of about 1% exists, whether CPU pinning changes it, and what caused a 27% drift between two later sessions remain inconclusive. What did not hold traced to four pitfalls: kernel reclaim of zero-filled memory, a load client that measured socket exhaustion, a sampling window longer than the load, and absolute numbers that do not travel across software environments. Each has a cheap check (Table I). The raw data, scripts and a one-command rebuild of every table and figure are published at https://github.com/vinayaktyagi10/cicd-monitoring-study.

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